The DRanking Protocol: Verifiable Distributed Ranking and Sybil-Bounded Admission for Structured Peer-to-Peer Networks

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Abstract

DRanking converts locally observed peer behavior in an open structured peer-to-peer network into a globally agreed, verifiable service weight that gates admission to protocol roles. Identity creation is free, so no mechanism bounds adversarial count [1]; the protocol instead bounds adversarial weight and prices it. The instruments are: counterparty-signed service receipts in place of unsigned counters; a seed-anchored trust flow $t_{k+1} = \alpha C_k^{\top} t_k + (1-\alpha)s$ obeying conservation, attenuation, and revocation laws, with a proved bound $T_k(A) \leq \alpha g/(1-\alpha)$ on Sybil mass; trust-weighted verifiable random sortition in place of oracle sampling; a proof-of-authority ledger trusted for ordering only, with equivocation proofs as first-class slash objects; and a circuit-friendly epoch transition folded by Nova-family incrementally verifiable computation, giving constant-cost verification to browser nodes. Rank is soulbound and slashable; reward is a separate transferable asset; the authority-to-stake transition is conditioned on a security-budget crossover.

I. Introduction

The 2023 draft of this protocol scored behavior per identity. Identities are free; every component secured against lying nodes fell to numerous nodes. This rewrite keeps the draft's decomposition — local evidence, global aggregation, sampling, ledger, incentives — and replaces the quantity being secured:

$$
\begin{aligned}
\mathsf{cost}(\mathsf{did}) &= 0,\\
|\mathsf{Adv} \cap V| &\leq \infty,\\
\mathsf{secure}(|\cdot|) &= \bot \quad [1],\\
\mathsf{secure} &:= \mathsf{secure}(T(\cdot)),\\
T(A) &= \textstyle\sum_{v\in A} t(v),\\
\mathsf{claim} &= \mathsf{price}\big(T(\mathsf{Adv}) \geq \tfrac{1}{3}\big).
\end{aligned}
$$

Sections II–IX construct t, its evidence, its sampling, its ledger, and its economy; Section X tabulates the attack surface. Each mechanism is a term of price.

II. Model

Definition 1 (Substrate).

$$
\begin{aligned}
R &= \mathbb{Z}/2^{160}\mathbb{Z} \quad [2], [3],\\
\mathsf{did} &: K \to R, \quad \text{self-certifying},\\
\mathsf{link} &\subseteq V \times V, \quad \text{authenticated},\\
\mathsf{msg} &= \langle m, \sigma_{\mathsf{origin}} \rangle,\\
\mathsf{cost}(\mathsf{did}) &= 0,\\
\mathsf{cost}(\mathsf{link}) &= \varepsilon > 0.
\end{aligned}
$$

Definition 2 (Epoch).

$$
k \in \mathbb{N}, \qquad \mathsf{epoch}(k) = [\,\mathsf{fin}(\Sigma_k),\ \mathsf{fin}(\Sigma_{k+1})\,),
$$

delimited by ledger finality (Definition 10).

Assumption 1 (Adversary).

$$
\begin{aligned}
|\mathsf{Adv}| &\leq \infty,\\
\mathsf{collude},\ \mathsf{grind},\ \mathsf{patience} &\in \mathsf{Cap}(\mathsf{Adv}),\\
\mathsf{byzantine} &\in \mathsf{Cap}(\mathsf{Adv}),\\
\mathsf{forge}(\sigma_h),\ h \notin \mathsf{Adv} &\notin \mathsf{Cap}(\mathsf{Adv}),\\
\mathsf{break}(\mathsf{VRF}) &\notin \mathsf{Cap}(\mathsf{Adv}).
\end{aligned}
$$

Assumption 2 (Honest weight).

$$
T_k(\mathsf{Adv}) \ < \ \tfrac{1}{3}\, T_k(V) \qquad \forall k.
$$

No bound is assumed on $|\mathsf{Adv}|$.

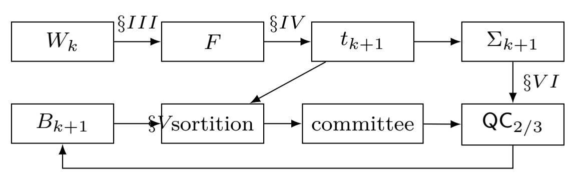

Figure 1. Epoch cycle: receipts fold into trust, trust seats the committee, the committee finalizes the ledger, finality seeds the beacon.

III. Service Receipts

Unsigned counters are first-person testimony; aggregated, they are hearsay. The unit of evidence is a countersigned receipt.

Definition 3 (Receipt).

$$
\rho = \langle \mathsf{srv}, a, b, k, n, \sigma_b \rangle,
$$

$$
\begin{aligned}
\mathsf{srv} &\in \{\mathsf{relay}, \mathsf{store}, \mathsf{probe}\},\\
a &= \text{server}, \quad b = \text{signer/served},\\
\mathsf{valid}(\rho) &\equiv \mathsf{Vrfy}(k_b, \sigma_b, \mathsf{srv}\|a\|b\|k\|n) = \top\\
&\quad \wedge\ (b, k, n)\ \text{fresh}.
\end{aligned}
$$

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Proposition 1 (Forgery surface).

$$\mathsf{valid}(\rho) \wedge b \notin \mathsf{Adv} \;\Rightarrow\; \rho \text{ issued by } b.$$

Proof. Immediate from $\mathsf{forge}(\sigma_b) \notin \mathsf{Cap}(\mathsf{Adv})$ (Assumption 1). Manufacturable receipts are exactly those with $b \in \mathsf{Adv}$ (self-dealing); these are neutralized by signer-trust weighting (Section IV). □

Definition 4 (Receipt graph).

$$\begin{aligned} \mathsf{val}(\rho) &= \mu_{\mathsf{srv}(\rho)} \cdot \mathsf{units}(\rho), \quad \mu \text{ published per kind},\\ W_k &\in \mathbb{R}^{V\times V}_{\geq 0},\\ W_k[b,a] &= \textstyle\sum\{\mathsf{val}(\rho) : \rho \text{ valid},\ \mathsf{sig}\ b,\ \mathsf{srv}\ a,\ \mathsf{ep}\ k\},\\ W_k[b,\cdot]\,\mathbf{1} &\leq w_{\max}. \end{aligned}$$

The cap makes countersigning a budgeted act; the protocol certifies performed service, never synthetic load (cf. Constraint 1).

IV. Trust Algebra

Definition 5 (Seed vector).

$$\begin{aligned} s &\in [0,1]^V, \qquad \textstyle\sum_v s_v = 1,\\ \mathsf{supp}(s) &= S \quad \text{(authority set)}. \end{aligned}$$

Law 1 (Conservation).

$$C_k[b,a] = \frac{W_k[b,a]}{\max\bigl(w_{\max},\ W_k[b,\cdot]\,\mathbf{1}\bigr)}, \qquad C_k[b,\cdot]\,\mathbf{1} \leq 1.$$

Law 2 (Attenuation).

$$\alpha \in (0,1), \qquad \mathsf{flow}(\ell \text{ hops}) \leq \alpha^{\ell}.$$

Law 3 (Revocation).

$$\begin{aligned} \mathsf{slash}(v) &\Rightarrow t(v) := 0,\ v \notin \mathsf{carrier}(t),\\ t_{k+1} &= f(W_k, t_k) \quad \text{(re-derived per epoch)},\\ s &\quad \text{withdrawable per seed.} \end{aligned}$$

Definition 6 (Trust update).

$$t_{k+1} = \alpha\, C_k^{\mathsf{T}} t_k + (1-\alpha)\, s. \tag{1}$$

One power-iteration step per epoch toward $t^* = (1-\alpha)(I-\alpha C^{\mathsf{T}})^{-1}s$ [4], [5]; the fixpoint is approached along chain time, never computed within an epoch (cf. Definition 12).

Invariant 1 (Mass).

$$\textstyle\sum_v t_0(v) = 1 \;\Rightarrow\; \sum_v t_k(v) \leq 1 \quad \forall k.$$

Proof.

$$\begin{aligned} \textstyle\sum_v t_{k+1}(v) &= \alpha\,\mathbf{1}^{\mathsf{T}} C_k^{\mathsf{T}} t_k + (1-\alpha)\\ &\leq \alpha\,\mathbf{1}^{\mathsf{T}} t_k + (1-\alpha) \quad \text{(Law 1)}\\ &\leq \alpha + (1-\alpha) = 1 \quad \text{(induction)}. \end{aligned}$$

□

Proposition 2 (Sybil influence bound). Partition $V = H \uplus A$, $S \subseteq H$. Let

$$g_k = \sum_{b\in H,\, a\in A} C_k[b,a]\, t_k(b), \qquad g = \sup_k g_k.$$

Then

$$T_k(A) = \sum_{v\in A} t_k(v) \;\leq\; \frac{\alpha\, g}{1-\alpha} \qquad \forall k.$$

Proof. $S \cap A = \varnothing$ kills the seed term over A. Splitting (1) by origin region and applying Law 1 to internal circulation,

$$T_{k+1}(A) \leq \alpha\bigl(g_k + T_k(A)\bigr).$$

From $T_0(A) = 0$,

$$T_k(A) \leq \alpha g \sum_{i\geq 0} \alpha^i = \frac{\alpha g}{1-\alpha}.$$

□

$T(A)$ is linear in honest endorsement granted (g), independent of $|A|$. Defense concentrates on g: endorsement spends budget (Law 1), and detection claws it back (Law 3).

Proposition 3 (Farming price). Instantiating Proposition 2 at $A = \{v\}$: sustaining weight w requires per-epoch honest endorsement

$$g_v \;\geq\; \frac{1-\alpha}{\alpha}\, w \qquad \text{for every epoch the weight is held,}$$

so

$$\mathsf{farm}(w, \ell \text{ epochs}) \;\geq\; \ell \cdot \mathsf{cost}\Bigl(\tfrac{1-\alpha}{\alpha}\, w\Bigr) \;+\; \Pr[\mathsf{detect}] \cdot w,$$

the second term by Law 3: one accepted equivocation forfeits the stock. This is the farm function of Proposition 9, made explicit.

Proof. $T_k(\{v\}) \leq \alpha g_v/(1-\alpha)$; rearrange, multiply by holding duration, add the slash expectation. □

Proposition 4 (Geometric convergence). For stationary $C_k = C$, the map $\Phi(x) = \alpha C^{\mathsf{T}} x + (1-\alpha)s$ satisfies

$$\|\Phi(x) - \Phi(y)\|_1 \leq \alpha \|x-y\|_1,$$

hence t^* is the unique fixpoint and

$$\|t_k - t^*\|_1 \;\leq\; \alpha^k\, \|t_0 - t^*\|_1.$$

Proof.

$$\begin{aligned} \|C^{\mathsf{T}}(x-y)\|_1 &= \textstyle\sum_a \bigl|\sum_b C[b,a](x_b - y_b)\bigr|\\ &\leq \textstyle\sum_b \bigl(\sum_a C[b,a]\bigr)|x_b - y_b| \leq \|x-y\|_1 \end{aligned}$$

by Law 1; Banach on the affine contraction. Under drifting C_k the iterate tracks the moving fixpoint with lag $O(\alpha^j)$ per j epochs of stability. □

Definition 7 (Soulbound rank).

$$\begin{aligned} \mathsf{rank}_k(v) &= t_k(v),\\ \mathsf{transfer} &\notin \mathsf{Ops}(\mathsf{rank}),\\ \mathsf{rank} &\xrightarrow{\mathsf{slash}} 0,\\ \mathsf{rank} &\xrightarrow{\mathsf{epoch}} \text{re-derived (1)}. \end{aligned}$$

(Cf. Proposition 9.)

Definition 8 (Dilution schedule).

$$\begin{aligned} \mathsf{vol}_k &= \textstyle\sum_{i\leq k} \mathbf{1}^{\mathsf{T}} W_i\, \mathbf{1},\\ 1-\alpha_k &= D(\mathsf{vol}_k), \quad D \text{ non-increasing, published},\\ \bar\alpha &= \sup_k \alpha_k \;<\; 1,\\ t_0 &= s \quad \text{(cold start)}. \end{aligned}$$

Propositions 2–4 hold under the time-varying schedule with $\alpha := \bar{\alpha}$.

Definition 9 (Threshold authority).

$$\begin{aligned}\mathsf{Ops}(S) &= \{\mathsf{withdraw}, \mathsf{rotate}, \mathsf{endorse}_0\},\\ o \in \mathsf{Ops}(S) &\Rightarrow \mathsf{QC}_\theta(S)(o), \quad \theta \geq \tfrac{2}{3},\\ |\mathsf{compromised}| < \theta|S| &\Rightarrow \mathsf{Cap}(\mathsf{Adv}) \cap \mathsf{Ops}(S) = \varnothing.\end{aligned}$$

V. Sortition

Proposition 5 (Statistical undetectability). Model $\mathsf{did} = \mathsf{h}(\mathsf{pk})$, h a random oracle. For the adversarial placement strategy "sample fresh keys," honest and adversarial positions are identically distributed on R:

$$\Delta_{\mathrm{TV}}\big(\mathsf{pos}(H \uplus A),\ \mathsf{pos}(H')\big) = 0 \quad \forall\,|A|,$$

hence every position-based test (entropy, Kolmogorov–Smirnov) has distinguishing advantage 0 against Sybil density.

Proof. $\mathsf{h}(\cdot)$ uniform on R for every key, honest or adversarial; a union of i.i.d. uniforms is distribution-invariant in its partition. □

Placement tests detect biased sampling; the threat is cheap identity. Validation must be cryptographic and economic, and a global random seed is a bootstrap circularity (a randomness beacon presupposes the committee being selected). Both are removed by sortition [6], [7].

Algorithm 1 Trust-weighted sortition (node v, epoch k, role r)

$B_k \leftarrow \mathsf{hash}(\mathsf{fin}(\Sigma_k))$ ▷ beacon: fixed by finality
$(y, \pi) \leftarrow \mathsf{VRF}_{\mathsf{sk}_v}(B_k \| r)$
if $y/2^{|y|} < \tau \cdot t_k(v)$ **then**
  v holds r in epoch k; publishes π

Proposition 6 (Grinding closure).

$$\left.\begin{aligned}\mathsf{sk}_v \text{ fixed} &\prec B_k \text{ exists},\\ B_k \text{ fixed} &\prec \mathsf{Adv} \text{ acts},\end{aligned}\right. \Rightarrow \mathsf{grind} = \bot.$$

Proof. Both VRF inputs are committed before either party can condition on the output; unpredictability from Assumption 1. □

Proposition 7 (Committee safety). Let seats be independent indicators with $\mathbb{E}[X] = \tau$, $\mathbb{E}[X_A] = \tau\, T(A)$, $T(A) < \frac{1}{3}$. By Chernoff–Hoeffding,

$$\Pr\Big[X_A \geq \tfrac{\tau}{3}\Big] \;\leq\; \exp\Big(-\tau\, \mathrm{D}\big(\tfrac{1}{3} \,\|\, T(A)\big)\Big),$$

with D the binary relative entropy: committee dishonesty decays exponentially in committee size. Sortition is the transport of Assumption 2 from weight to seats; no reported quantity exists for a coalition to steer, so the draft's median game is vacuous by construction.

VI. Admission Ledger

Definition 10 (Ledger).

$$\begin{aligned}\Sigma_k &= \langle \mathsf{rt}(t_k), \mathsf{rt}(V_k), \mathsf{rt}(\mathsf{sl}_k), \mathsf{rt}(\mathsf{bal}_k), \mathsf{ver}\rangle,\\ \mathsf{fin}(\Sigma_k) &\equiv \mathsf{QC}_{2/3}(S_k) \quad [8], [9],\\ \mathsf{content} &= \{\mathsf{rt}(W_k)\} \cup \mathsf{sl} \cup \Delta V \cup \mathsf{issue},\\ \mathsf{data}(W_k) &\in \mathsf{Overlay\ storage}\ \text{(not on chain)}.\end{aligned}$$

Obligation 1 (Minimal trust).

$$\begin{aligned}\forall o \in \mathsf{content}: \ \mathsf{Verify}(o) &\perp \mathsf{honest}(S),\\ \mathsf{Cap}(\mathsf{quorum}) &= \{\mathsf{order}, \mathsf{censor}\},\\ \mathsf{forgeable}(\mathsf{quorum}) \ni o &\Rightarrow \text{design defect.}\end{aligned}$$

Definition 11 (Equivocation proof).

$$e = \langle m_1, \sigma_v(m_1), m_2, \sigma_v(m_2)\rangle,$$

$$\mathsf{slot}(m_1) = \mathsf{slot}(m_2) \ \wedge\ m_1 \neq m_2 \ \Rightarrow\ \mathsf{slash}(v),$$

verification assumption-free; e is a first-class ledger object, submittable by anyone.

Obligation 2 (Force inclusion).

$$\mathsf{ack}_S(x) \ \wedge\ x \notin \Sigma_{k+\Delta} \ \Rightarrow\ \mathsf{ack}_S(x) \in \mathsf{sl}\ \text{(quorum slash)},$$

bounding censorship, the residual validator power, by latency Δ.

Equivocation defeats patience: months of farmed standing forfeit on one contradiction, so $\mathbb{E}[\mathsf{farm}] \downarrow$ with detection probability.

VII. Folded State

Browser wasm nodes can neither replay history nor audit a validator signature; the chain must be succinct [10].

Definition 12 (Epoch transition).

$$F(\Sigma_k, \mathsf{rt}(W_k)) = \Sigma_{k+1},$$

Algorithm 2 F — pure, deterministic, per epoch

1: $\mathsf{sl}_{k+1} \leftarrow \mathsf{sl}_k \cup \{\mathsf{verify}(e) : e \text{ submitted}\}$
2: $t' \leftarrow t_k \restriction_{V \setminus \mathsf{sl}_{k+1}}$ ▷ Law 3
3: $t_{k+1} \leftarrow \alpha_k\, C_k^{\mathsf{T}} t' + (1 - \alpha_k)\, s$ ▷ Eq. (1), Def. 8
4: $\mathsf{bal}_{k+1} \leftarrow \mathsf{bal}_k + \mathsf{issue}(W_k, t_k) - \mathsf{burn}(\mathsf{fees}_k)$
5: $V_{k+1} \leftarrow \mathsf{vset}(\Sigma_{k+1})$ ▷ Obligation 7
6: **return** $\langle \mathsf{rt}(t_{k+1}), \mathsf{rt}(V_{k+1}), \mathsf{rt}(\mathsf{sl}_{k+1}), \mathsf{rt}(\mathsf{bal}_{k+1}), \mathsf{ver}\rangle$

$\mathsf{cost}(F) = O(\mathsf{nnz}(W_k))$ adds/mults over committed data.

Line 3 is why the fixpoint is unrolled: one sparse matrix–vector product arithmetizes trivially; an eigenvector solve does not.

Definition 13 (Folding). Nova-family IVC [11]:

$$\begin{aligned}\pi_k &\vdash \Sigma_0 \xrightarrow{F^k} \Sigma_k,\\ |\pi_k| &= O(1), \qquad \mathsf{Verify}(\pi_k) = O(1),\\ \mathsf{sync} &= \mathsf{Verify}(\pi_k) \wedge \mathsf{QC}_{2/3}(\Sigma_k),\\ \mathsf{prove} &\parallel \mathsf{consensus} \quad \text{(asynchronous, trails head).}\end{aligned}$$

Proposition 8 (Linearity).

$$\mathsf{fin}(\Sigma_k) \text{ instant (BFT)} \;\Rightarrow\; \mathsf{chain} \cong (\Sigma_0, \Sigma_1, \dots) \text{ linear},$$

hence IVC applies with no re-folding branch; probabilistic-finality chains admit no such reduction.

Proof. Quorum intersection: two finalized blocks at one height require $\geq \frac{1}{3}$ of the finality committee to equivocate (Definition 11) — contradicting Definition 9 in the authority phase, and Proposition 7 with Assumption 2 in the staked phase. □

Obligation 3 (Circuit discipline).

$$\begin{aligned} &\forall \text{ extension } F' : \\ &\qquad F' \text{ pure} \quad \wedge \quad F' \in \mathsf{CheapArith}(\mathsf{committed}). \end{aligned}$$

The ledger is an admission ledger, not a contract platform.

Obligation 4 (Upgrade versioning).

$$\begin{array}{rcl} F \mapsto F' & \Rightarrow & \mathsf{vk} \mapsto \mathsf{vk}', \\ \mathsf{ver} & \in & \Sigma, \\ \mathsf{bind}(\mathsf{vk}, \mathsf{vk}') & \text{at} & \text{committed epoch;} \end{array}$$

shipped at genesis, not retrofitted.

Folding provides validity only: $\mathsf{canonicity} \in \mathsf{consensus}$, $\mathsf{availability} \in \mathsf{overlay}$.

Definition 14 (Commitment granularity).

$$\begin{array}{rcl} \Gamma & \in & \{\mathsf{full}, \mathsf{agg}, \mathsf{blind}\}, \\ \mathsf{leak}(\mathsf{full}) & = & \{(b, a, W[b,a])\} \quad \text{(the traffic graph)}, \\ \mathsf{leak}(\mathsf{agg}) & = & \{(v,\ \mathsf{deg}^{\pm}(v))\}, \\ \mathsf{leak}(\mathsf{blind}) & = & \varnothing, \quad F \text{ consumes ZK openings}, \\ \mathsf{replay}(\Gamma) & = & \top \iff \Gamma = \mathsf{full}. \end{array}$$

Γ trades public replayability against surveillance of the service graph; the choice is a standing design decision, and $\Gamma = \mathsf{blind}$ prices its privacy in circuit cost (Obligation 3).

VIII. State Relation

The ledger state machine in TLA form [12], the model-checking surface for finite instances [13]:

$$\begin{array}{rcl} \mathsf{Spec} & \triangleq & \mathsf{Init} \wedge \Box[\mathsf{Next}]_\Sigma, \\ \mathsf{Init} & \triangleq & \Sigma = \langle \mathsf{rt}(s), \mathsf{rt}(S), \varnothing, \mathsf{rt}(\mathbf{0}), \mathsf{ver}_0 \rangle, \\ \mathsf{Next} & \triangleq & \exists\, W :\ \mathsf{QC}_{2/3}(\Sigma') \ \wedge\ \Sigma' = F(\Sigma, \mathsf{rt}(W)). \end{array}$$

Invariant 2 (Slash monotone).

$$\mathsf{sl} \subseteq \mathsf{sl}', \qquad v \in \mathsf{sl} \Rightarrow t(v) = 0.$$

Proof. Lines 1–2 of Algorithm 2: sl only grows by union; the restriction $t\restriction_{V \setminus \mathsf{sl}}$ zeroes slashed carriers before every update, and Equation (1) grants no mass to $v \notin \mathsf{carrier}$. □

Invariant 3 (Version monotone).

$$\mathsf{ver}' \geq \mathsf{ver},$$

by Obligation 4: vk rebinding occurs only at committed epoch boundaries, forward.

Invariant 4 (Unique chain).

$$\mathsf{fin}(\Sigma_k) \wedge \mathsf{fin}(\Sigma_k') \;\Rightarrow\; \Sigma_k = \Sigma_k',$$

by Proposition 8 (quorum intersection under Assumption 2).

Obligation 5 (Checkable instances). Every implementation change to F ships with a finite-instance model ($|V| \leq n_0$, bounded epochs) checking Invariants 1, 2–4 before merge.

IX. Economics

Definition 15 (Two assets).

$$\begin{array}{rcl} \mathsf{rank} & : & \text{soulbound (Def. 7),} \\ & & \text{gates admission/sortition,} \\ \mathsf{reward} & : & \text{transferable,} \\ \mathsf{issue}_k(v) & \propto & \big(W_k^{\mathsf{T}} t_k\big)(v). \end{array}$$

Proposition 9 (Payable rank defeats cost anchoring). Let rank be transferable at price p. Then

$$\begin{array}{rcl} \mathsf{cost}_{\mathsf{Adv}}(m) & = & \min\big(\mathsf{farm}(m),\ p \cdot m\big), \\ u_{\mathsf{exit}}(\mathsf{hold}) = 0 \ \wedge\ p > 0 & \Rightarrow & \text{exiting nodes sell,} \\ \text{supply} > 0 & \Rightarrow & p \leq \inf_v \mathsf{farm}_v(m)/m, \end{array}$$

so acquisition cost decouples from behavior. Non-transferability removes the second branch of the min, restoring $\mathsf{farm}(m) = \text{service} + \text{time} + \text{slash risk}$ as the sole path. □

Invariant 5 (Balance conservation).

$$\mathbf{1}^{\mathsf{T}} \mathsf{bal}_{k+1} = \mathbf{1}^{\mathsf{T}} \mathsf{bal}_k + \mathsf{issue}_k - \mathsf{burn}_k,$$

by line 4 of Algorithm 2: no other term touches bal; supply changes only through published issuance and burn.

Definition 16 (Fee loop).

$$\begin{array}{rcl} \mathsf{Sinks} & = & \{\mathsf{store}_{>\rho},\ \mathsf{QoS},\ \mathsf{namespace}\}, \\ \mathsf{fees} & \to & \mathsf{burn} \ \cup\ \mathsf{issue\ pool}, \\ \mathsf{anchor}_{\mathsf{PoS}} & \leq & f(\mathsf{demand}(\mathsf{Sinks})). \end{array}$$

Obligation 6 (Security-budget crossover).

$$\mathsf{activate}(\mathsf{PoS}) \iff \mathsf{Bond} \ \geq\ \beta \cdot \widehat{\mathsf{payoff}}(\mathsf{attack}), \quad \beta > 1,$$

conditioned, never scheduled; pre-crossover, PoA finality overlays PoS production, weighted down by $D(\mathsf{vol}_k)$ (Definition 8).

Obligation 7 (Validator-set abstraction).

$$\begin{array}{rcl} \mathsf{vset} & : & \Sigma \to 2^V, \\ \mathsf{vset}_{\mathsf{PoA}} & = & \mathsf{const}\ S, \\ \mathsf{vset}_{\mathsf{PoS}} & = & \mathsf{sortition}(\mathsf{Bond}); \end{array}$$

the transition changes one pure function, not the engine.

Constraint 1 (Rejected anchors).

$$\begin{array}{rcl} \mathsf{PoW} & : & \mathsf{cost}(51\%) = \mathsf{rent}(\mathsf{hash}) \cdot O(\text{hours}) \ \Rightarrow\ \bot, \\ \mathsf{PoT} & : & W_{\mathsf{synth}}[b,a],\ a, b \in \mathsf{Adv} \ \not\perp\ W_{\mathsf{real}} \\ & & \Rightarrow\ \text{bandwidth burned},\ \Delta T(\mathsf{Adv}) \approx 0 \ \Rightarrow\ \bot. \end{array}$$

Receipted real service with trust weighting is proof-of-traffic minus the synthetic load.

X. Security Analysis

Table I
Threat matrix

Attack	Bound	Residual
Sybil flood	Prop. 2	$\alpha g/(1-\alpha)$
Self-dealing	Prop. 1 + signer-trust weight; cap (Def. 4)	zero-trust circulation
Patient farming	Laws 1–3; total slash (Def. 11)	undetected deviation
Whitewashing	fresh DID $\Rightarrow t = 0$; slow earn, instant loss	re-entry at zero
Grind/eclipse	Prop. 6	—
Censorship	Obl. 2	latency Δ
Seed compromise	threshold S; revocation; Def. 8	pre-dilution window
Rank market	Prop. 9	out-of-band key sale

$$\begin{aligned} \mathsf{sale}(\mathsf{sk}_v) &\Rightarrow \text{buyer inherits slash exposure,} \\ \mathsf{PoA} &= \text{trust root until Obl. 6.} \end{aligned}$$

XI. Related Work

Local scoring descends from eDonkey queues [14] and BitTorrent choking [15]; propagation from EigenTrust and personalized PageRank [4], [5], here unrolled along epochs over signed evidence; the attack-edge bound pattern from SybilGuard/SybilLimit/SybilRank [16]–[18], instantiated over a service graph. Brahms [19] addresses adversarial sampling that the draft's tests could not (Proposition 5); DRanking sidesteps via Algorand-style sortition [6], [7] weighted by trust. S/Kademlia [20] prices identifier creation (complementary one-time cost). Finality follows PBFT lineage [8], [9]; succinctness follows Mina [10] via Nova folding [11]; impossibility framing is Douceur's [1]; the external-cost anchor observation is Nakamoto's [21].

XII. Conclusion

$$\mathsf{safe} \iff \mathsf{cost}\big(T(\mathsf{Adv}) \geq \tfrac{1}{3}\big) > \widehat{\mathsf{payoff}}(\mathsf{attack}),$$

and every mechanism above — receipts, the trust laws, sortition, the ledger, folding, the two-asset economy — is a term on the left-hand side.

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